



FairMarket-RL: LLM-driven fairness shaping in multi-agent energy trading for smart distribution grids

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ARTICLE INFO

Keywords:

Fairness Shaping
Independent Proximal Policy Optimization (IPPO)
LLM-RL
Multi-Agent Reinforcement Learning (MARL)
Peer-to-Peer (P2P) Trading

ABSTRACT

Peer-to-peer (P2P) trading is increasingly recognized as a key mechanism for decentralized market regulation, yet existing approaches often lack robust frameworks to ensure fairness. This paper presents FairMarket-RL, a novel hybrid framework that combines Large Language Models (LLMs) with Reinforcement Learning (RL) to enable fairness-aware trading agents. In a simulated P2P microgrid with multiple sellers and buyers, the LLM acts as a real-time fairness critic, evaluating each trading episode using two metrics: Fairness-To-Buyer (FTB) and Fairness-Between-Sellers (FBS). These fairness scores are integrated into agent rewards through scheduled λ -coefficients, forming an adaptive LLM-guided reward shaping loop that replaces brittle, rule-based fairness constraints. A linear λ ramping curriculum stabilizes training and gradually incorporates fairness with no loss in convergence. Agents are trained using Independent Proximal Policy Optimization (IPPO) and achieve equitable outcomes, fulfilling over 90% of buyer demand, maintaining fair seller margins, and consistently reaching FTB and FBS scores above 0.80. Notably, the framework demonstrates resilience against adversarial behaviors and noise, maintaining stable performance even with corrupted fairness inputs. The training process demonstrates that fairness feedback improves convergence, reduces buyer shortfalls, and narrows profit disparities between sellers. Because the language-based critic requires no hand-crafted fairness features, the framework is structured to extend to settings with larger numbers of prosumers, a direction we examine qualitatively and leave to future work for full validation. FairMarket-RL thus offers an equity-driven approach for autonomous trading in decentralized energy systems.

1. Introduction

Peer-to-peer energy trading on distribution feeders lets prosumers transact locally without a central market operator, boosting autonomy and resilience. However, this decentralization also creates equity risks when dominant sellers inflate prices or certain buyers consistently absorb excessive costs (Zhou et al., 2018; Tushar et al., 2020). Recent Multi-agent reinforcement-learning (MARL) prototypes now clear local energy bids while respecting grid constraints yet pronounced payoff disparities persist; Chen and Liu's networked-MARL testbed, for example, required ad-hoc price caps to stay socially acceptable (Chen and Liu, 2023). Classical “fairness patches” such as potential-based shaping, proportional-fair scoring, and profit-variance penalties tame extreme outcomes only in static conditions and deteriorate as load,

generation, or topology shift (Tushar et al., 2018). This underscores the enduring challenge of achieving fair outcomes in P2P markets.

More principled solutions recast equity as an explicit objective in the learning process. For instance, Siddique et al. optimized a lexicographic maximin welfare, and Zimmer et al. introduced an auxiliary equity network, both yielding roughly 20% fairness improvements at the cost of bespoke metrics and fragile hyper-parameter tuning (Siddique et al., 2020). Notably, these studies illustrate a known fairness efficiency trade-off: naive fairness enhancements can reduce total welfare (Bhat et al., 2021; Rahmattalabi et al.). In contrast, the emergence of instruction tuned LLMs offers a more adaptive path. Modern LLMs absorb broad ethical and economic priors through human feedback training (Ouyang et al., 2022), enabling them to reason about concepts like justice and equity in free-form language. Such models can ingest structured ledgers

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<https://doi.org/10.1016/j.tej.2026.107548>

Received 7 July 2025; Received in revised form 16 May 2026; Accepted 26 May 2026

Available online 2 June 2026

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of prices, unmet demand, and profits, then return compact numeric fairness scores without hand-crafted features or domain-specific tuning (Silva et al., 2025). We posit that an LLM can serve as a dynamic “fairness critic” inside the MARL loop, providing interpretable numeric feedback that replaces brittle rule-based fairness rewards.

Building on the stability of IPPO (Schroeder de Witt et al., 2020), we embed a pre-trained LLM as a real-time fairness critic in a decentralized MARL framework. IPPO is chosen for its strong performance in multi-agent settings, it can match or outperform more centralized algorithms on complex tasks (Schroeder de Witt et al., 2020), making it a robust baseline for independent learning. After each trading episode, the LLM observes the collective outcome and outputs two fairness metrics – FTB for each buyer and FBS overall which modulate the agents’ economic rewards through scheduled time-varying λ -coefficients. We gradually increase the fairness weight λ during training a form of curriculum to smoothly incorporate fairness objectives without destabilizing convergence. This plug-and-play feedback loop closes the gap left by monetary incentives alone, directly addressing the fairness efficiency tension highlighted in recent MARL surveys (Bhat et al., 2021; Rahmattalabi et al) and aligning with emerging “Constitutional AI” principles (Bai). The resulting framework, FairMarket-RL, achieves both equitable and efficient outcomes in a simulated P2P microgrid market. In our case study, learning agents meet over 90% of total demand and maintain fair profit margins of roughly 20–30% for all sellers, while achieving fairness scores above 0.85 for both FTB and FBS. We also find that LLM-guided agents avoid exploitative behaviors like price gouging or hoarding, demonstrating robustness even under noisy or adversarial LLM inputs by discarding invalid feedback. By integrating a language-model-based fairness judge into decentralized trading, FairMarket-RL offers a promising route toward socially aligned P2P electricity markets. Moreover, our LLM-centric approach is complementary to privacy-preserving frameworks, for instance, Singh and Singh (Singh and Singh, 2025) use federated multi-agent learning to optimize P2P pricing without centralizing data. FairMarket-RL’s fairness evaluator could be deployed in a federated or on premises fashion alongside such methods, ensuring equity and privacy in real-world smart grids.

The remainder of this paper is organized as follows: Section 2 describes the system architecture of FairMarket-RL, detailing the decentralized multi-agent framework, two-stage trading interaction, and the integration of the LLM fairness critic within the training loop and inference pipeline. Section 3 presents the mathematical modeling of the P2P trading environment, formalizing the multi-agent game structure, global state representation, episode dynamics, and raw reward formulations for sellers and buyers. Section 4 combines our learning methodology and experimental results, beginning with algorithm selection (IPPO versus MAPPO, MADDPG, and QMIX baselines), followed by case-study environment parameters, fairness integration via curriculum learning, learning dynamics analysis, key performance indicators, ablation studies on early-to-mid training behavior, fairness-efficiency trade-off analysis, and robustness evaluations under adversarial conditions. Section 5 explores potential applications of the framework in power and energy systems, emphasizing scalability to community microgrids and large-scale smart grids with distributed prosumers. Finally, Section 6 concludes with key findings, implications for real-world deployment, and future research directions for enhancing fairness, privacy, and interpretability in autonomous energy markets.

2. System architecture

FairMarket-RL is a decentralized multi-agent framework in which (N_S) seller agents and (N_B) buyer agents interact through a shared market environment. Each episode unfolds in a two-stage trading interaction: first, a seller offer stage where sellers ($S_1, \dots, S_{N_S}$) simultaneously post price–quantity offers based on their local states; second, a

buyer allocation stage where buyers ($B_1, \dots, B_{N_B}$) simultaneously allocate demand across the posted offers subject to budget and demand limits. After these two stages, the environment compiles an outcome ledger (offers, allocations, residual shortfall, realized payoffs). Fig. 1 illustrates the training loop with the LLM fairness path.

Seller and buyer policies are independent decentralized networks that output offers and allocations from local observations. The Prompt Skeleton serializes the episode’s outcome ledger into a deterministic JSON with fixed field order, units, and keys, and queries the LLM Fairness Critic. In this work the critic is an instruction-tuned OpenAI GPT-4o model, accessed through the chat-completions API and queried once per episode with the decoding temperature set to 0 so that an identical ledger yields identical fairness scores. The prompt has two fixed parts: a system message that defines the FTB and FBS criteria and instructs the model to return only a JSON object, followed by the serialized episode ledger supplied as the user message. The LLM returns FTB (a per-buyer vector) and FBS (a single market-level scalar). In the current system the critic is constrained to emit only these numeric values; it does not produce free-text explanations or rationales, which keeps the feedback compact and machine-parsable. Outputs pass through FTB/FBS decoders that perform type/range checks and normalization; episodes with invalid or unparseable responses are discarded to preserve a clean training signal. A Reward Shaping Module merges raw economic returns with the FTB/FBS signals, scaled by a λ -Scheduler that gradually increases the fairness weights, so agents first learn profitability and then internalize fairness. Trajectories with shaped returns are written to an on-policy Rollout Buffer, and an IPPO trainer updates each agent’s policy independently, an efficient choice for decentralized MARL with strong empirical performance (Schulman et al; Yu et al., 2022).

Within training, the per-episode flow is: (a) sellers post offers; (b) buyers allocate demand; (c) the environment summarizes the outcome and the Prompt Skeleton queries the LLM; (d) the Reward Shaping Module augments raw returns using FTB/FBS, scaled by time-varying $\lambda_{buy(t)}$ and $\lambda_{peer(t)}$; (e) shaped returns/advantages enter the rollout buffer and the IPPO trainer updates policies; (f) the process repeats with updated policies and fresh fairness feedback. Inference deployment uses the same seller-buyer interaction with the LLM and shaping path disabled learned policies act directly, keeping operation simple and low-latency. Fig. 2 illustrates the inference pipeline with the LLM shaping path disabled.

3. Mathematical modeling

We now present a formal description of the P2P trading environment, agents, and fairness metrics as a multi-agent game. The game is defined as a finite-horizon, turn-based game in (1):

$$G = \langle S, \{A^{S_i}\}_{i=1}^{N_S}, \{A^{B_j}\}_{j=1}^{N_B}, T, R \rangle \quad (1)$$

Where S is the state space, A^{S_i} and A^{B_j} are the action sets for seller S_i and buyer B_j , respectively, T is the deterministic transition function, and R is the reward vector. Each episode consists of two stages: first sellers act, then buyers act. The system can be represented by global state at time t is formulated as in (2):

$$s_t = [I^1, \dots, I^{N_S}, D^1, \dots, D^{N_B}, p^1, q^1, \dots, p^{N_S}, q^{N_S}, \sigma_t] \quad (2)$$

where I^i is the remaining inventory of seller S_i , and D^j is the unmet demand of buyer B_j . The tuples (p^i, q^i) are the active price–quantity offers of seller S_i from the seller stage. The flag $\sigma_t \in \{seller_{stage}, buyer_{stage}\}$ indicates whose turn it is. Initially, $\sigma_0 = seller_{stage}$ with all inventories I^i drawn from specified ranges and buyer demands D^j set similarly.

At $t = 0$, each seller S_i simultaneously chooses an action $(p_i, q_i) \in$

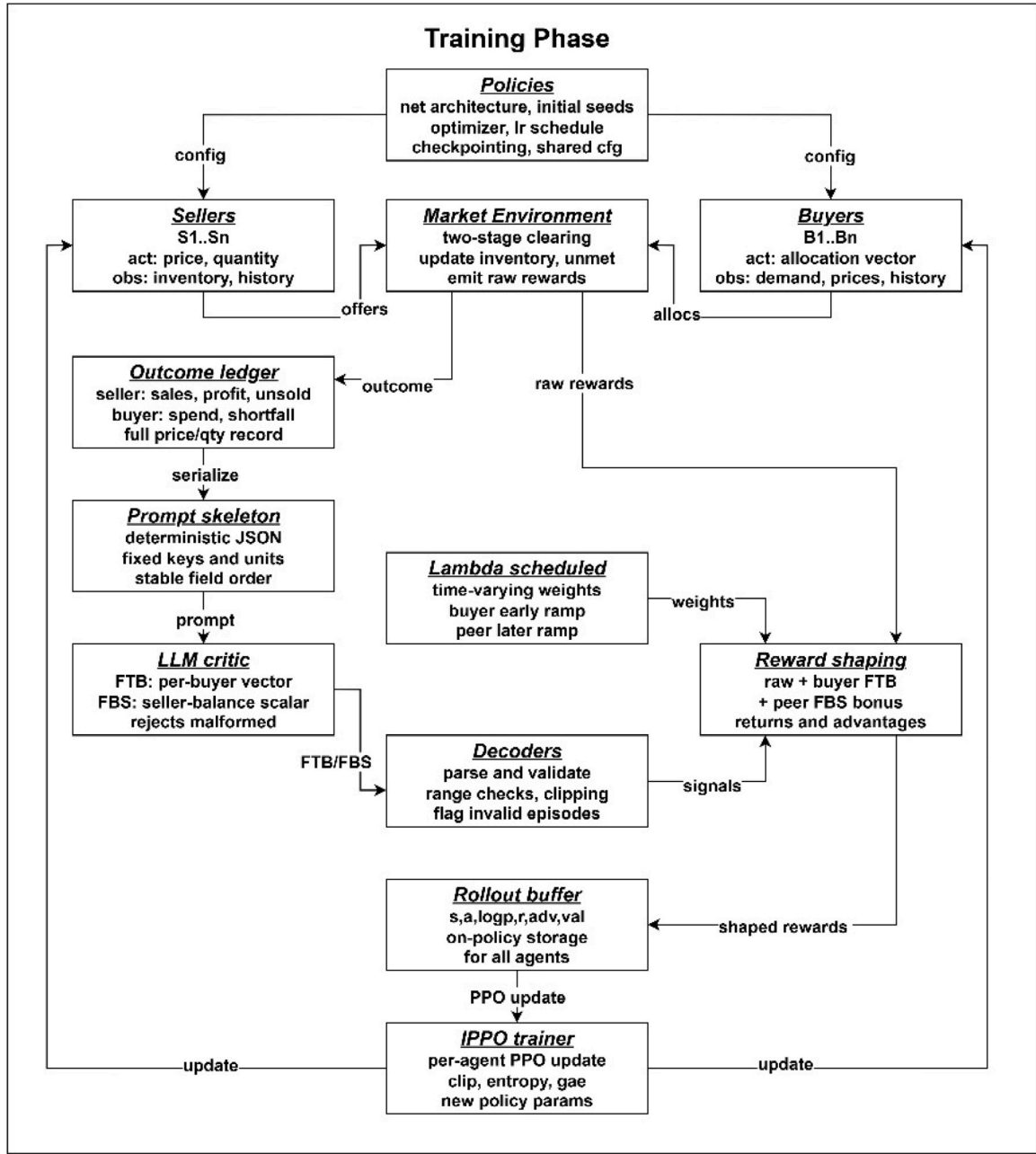


Fig. 1. Training architecture: sellers/buyers interact with the environment; the outcome is serialized to the LLM fairness critic, shaped via λ -weights, stored, and used by IPPO for policy updates.

$\{1, \dots, 10\} \times [0, I]$, posting a price p_i from a discrete price grid and a quantity q_i not exceeding its inventory. All seller offers take effect immediately. Once all sellers have posted, σ_t switches to *buyer_{stage}*. Now, each buyer B_j simultaneously chooses a demand allocation vector $b^{(j)} = (b_1^{(j)}, \dots, b_{N_s}^{(j)})$, where $b_i^{(j)}$ is the quantity B_j purchases from seller S_i . These allocations satisfy the budget and demand constraints as in (3):

$$\sum_{i=1}^{N_s} p_i b_i^{(j)} \leq \text{Budget}_j, \sum_{j=1}^{N_b} b_i^{(j)} \leq D_j, 0 \leq b_i^{(j)} \leq q_i \text{ for each } i \quad (3)$$

After all buyers have acted, the environment updates each seller's inventory and each buyer's unmet demand deterministically as in (4):

$$I_{\{new\}}^i = I_i - \sum_{j=1}^{N_b} b_i^{(j)}, D_{\{new\}}^j = D_j - \sum_{i=1}^{N_s} b_i^{(j)} \quad (4)$$

The episode ends once the buyer stage is complete. Any remaining unmet demand is pooled as total shortfall $D_{\{unsat\}} = \sum_{j=1}^{N_b} D_{\{new\}}^j$, and any unsold inventory per seller is $I_{\{unsold\}}^i = I_{\{new\}}^i$.

Before fairness shaping, each agent receives an economic reward based on trade outcomes as in (5), (6):

$$\text{Seller } S_i : r_{\{S_i\}}^{\{raw\}} = (p_i - c) \sum_{j=1}^{N_b} b_i^{(j)} - \alpha D_{\{unsat\}} - \beta I_{\{unsold\}}^i \quad (5)$$

Here c is the constant production cost per unit, so $(p_i - c)$ is seller i 's unit profit. The terms α and β are penalty coefficients: $\alpha D_{\{unsat\}}$ penalizes unmet total demand encouraging sellers to meet buyer demand, and $\beta I_{\{unsold\}}^i$ penalizes leftover inventory discouraging hoarding.

$$\text{Buyer } B_j : r_{\{B_j\}}^{\{raw\}} = - \sum_{i=1}^{N_s} p_i b_i^{(j)} - \alpha D_{\{unsat\}} \quad (6)$$

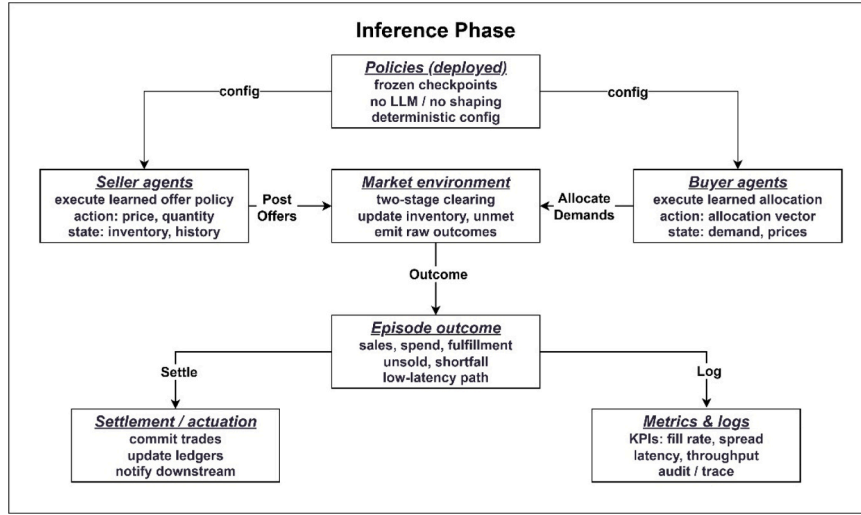


Fig. 2. Inference architecture: frozen seller/buyer policies act in the environment to settle trades with no LLM or shaping, yielding low-latency outcomes and logs.

The buyer's raw reward is negative of its spending, with the same shortfall penalty αD_{unsat} . We use a common shortfall penalty to align all agents to the goal of meeting overall demand.

To incorporate equity, we define two fairness scores via the LLM: a Fairness To Buyer vector $FTB = (FTB_1, \dots, FTB_{N_B})$ and a Fairness Between Sellers scalar FBS. Both FTB and FBS lie in $[0,1]$, where 1 indicates the ideal fair outcome. The LLM takes the final episode ledgers such as prices, allocations, unmet demand, profits, etc. and outputs these normalized scores. These scores then augment the agents' rewards with λ -weighted terms. Formally, the shaped rewards as in (7), (8):

$$\text{Seller : } R_{S_i} = r_{S_i}^{\text{raw}} + \lambda_{\text{buy}} W_B \left(\frac{1}{N_B} \sum_{j=1}^{N_B} FTB_j \right) + \lambda_{\text{peer}} W_P \left(\frac{\sum_j b_i^{(j)}}{\sum_{k,l} b_k^{(l)}} \right) FBS \quad (7)$$

$$\text{Buyer : } R_{\{B_j\}} = r_{\{B_j\}}^{\text{raw}} + \lambda_{\text{buy}} W_B FTB_j \quad (8)$$

In these equations, w_B and w_P are fixed coefficients tuning the strength of each incentive, and $\lambda_{\text{buy}(t)}$, $\lambda_{\text{peer}(t)}$ are scheduled weights that ramp from 0 to 1 over training time. By design, all fairness bonuses disappear when the λ -coefficients are zero initially, allowing agents to first learn the basic market dynamics and profit-making before fairness is emphasized. Any episode yielding an invalid or unparsable LLM output is discarded; this ensures the fairness feedback is always well-defined. If the LLM critic fails to return a valid FTB or FBS, we simply remove that episode from training updates to maintain a clean signal.

4. Learning methodology and experimental results

4.1. Algorithm selection: independent proximal policy optimization (IPPO)

We adopt an IPPO approach for multi-agent training, in lieu of more centralized or value-decomposition methods. In IPPO, each agent learns its own policy via PPO with only local observations and rewards, without a centralized critic, which keeps the learning process decentralized and scalable (Yu et al., 2021). This choice is justified by recent findings that decentralized PPO agents can achieve performance on par with or even superior to centralized algorithms in cooperative tasks (Wang et al., 2023). Indeed, our 5k-episode experiments (Fig. 3a and Fig. 3b) show that IPPO attains stable, high social welfare and fairness throughout training, whereas the baselines struggle to match this balance. The centralized Multi-Agent PPO (MAPPO) algorithm (Yu et al., 2021), which extends IPPO with a joint critic using global state information, reached competitive peak welfare in Fig. 3a, but fell short of IPPO on the fairness index in Fig. 3b. Off-policy alternatives fared worse: MADDPG (multi-agent DDPG (Lowe et al., 2017)) exhibited noisy learning curves and high variance, and the value-factorization method QMIX (Rashid et al., 2018) consistently underperformed in both fairness and efficiency. These observations align with prior benchmarks showing that MADDPG often underperforms other MARL algorithms on complex cooperative tasks (Samvelyan et al), and that QMIX and similar methods can fail to learn effectively under partial observability or sparse rewards (Samvelyan et al). In contrast, IPPO's on-policy training and

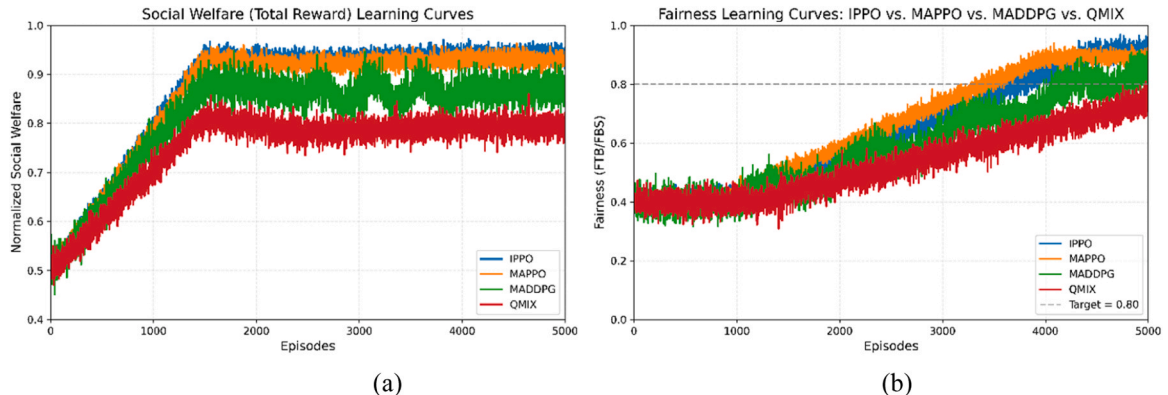


Fig. 3. (a) Learning curves over 5000 training episodes for IPPO and baselines showing social welfare (total reward); (b) corresponding fairness index trajectories.

independent value functions provide greater stability, which is crucial for learning a fairness-oriented policy. By avoiding centralized critics and off-policy complexities, IPPO better maintains training stability while optimizing the shared reward, yielding a favorable trade-off between individual learning simplicity and collective performance (Wang et al., 2023).

4.2. Case-study environment and training parameters

To ground the general multi-buyer, multi-seller framework in a concrete setting, we present a controlled case study with the non-trivial topology: two competing sellers and one budget-constrained buyer. Although modest, this configuration retains the core strategic tensions, price competition, inventory allocation, and buyer welfare that motivate our LLM-guided shaping mechanism. We emphasize that this case study is deliberately small. It isolates the essential market dynamics and keeps the curriculum and robustness analyses tractable, but it is not intended to demonstrate behavior at scale; validating the framework on larger networks with many sellers and buyers remains an important direction for future work, and the scalability discussion in Section 5 should be read in that light (Table 1).

4.3. Fairness integration via curriculum learning

To incorporate fairness objectives without destabilizing training, we employed a curriculum learning strategy via a linear λ -ramping schedule. Here, λ is the weight on the fairness-oriented reward term in each agent's reward function. During the full 20k-episode training run, λ was initialized at 0 and increased linearly to its maximum value over the course of training. This gradual schedule ensured the agents first learn to optimize efficiency social welfare and then progressively emphasize fairness, rather than confronting a hard fairness constraint from the start. As evidence of this design, Fig. 4 compares linear versus abrupt introduction of fairness weight. With a linear ramping of λ , agents achieved smooth and reliable fairness convergence, whereas an abrupt application of the full fairness weight caused slower and less stable improvements in the fairness metric. By the end of training, the curriculum had fully introduced the fairness objective without significant sacrifice to learning stability. This approach is motivated by the general principle that fairness efficiency trade-offs benefit from incremental adjustment (Hernandez-Leal et al). Imposing fairness too aggressively from the outset can reduce overall system performance or delay convergence, whereas a paced increase allows the agents to adjust their policies gradually (Hernandez-Leal et al). Our results confirm this: the agents maintained high welfare early on and then improved fairness as λ ramped up, avoiding the pitfalls of a static, abrupt fairness penalty.

4.4. Learning dynamics

Fig. 5a shows 500 episode moving-average returns for both sellers and the buyer, while Fig. 5b plots the LLM-driven fairness signals that shape those returns. In the profit-only warm-up ($\lambda = 0$) sellers raise prices at the buyer's expense. When buyer-fairness weight λ_{buy} ramps up (marker A, about episode 2000), sellers cut prices, buyer shortfalls shrink, and buyer returns rebound. A second shift occurs once the peer-

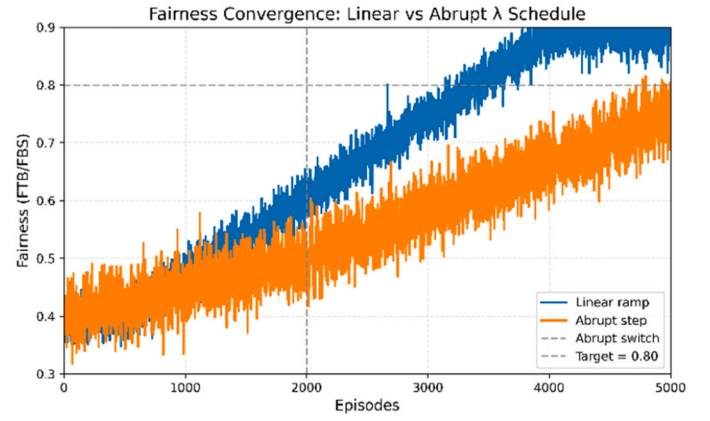


Fig. 4. A comparison of fairness convergence behavior under a linear increasing λ schedule versus an abrupt, static introduction of fairness weight.

fairness weight λ_{peer} continues to ramp up (marker B): the two seller profit curves gradually converge over the following episodes and are effectively merged by roughly episode 7500, as the peer-fairness signal equalizes their gains. FTB and FBS echo these shifts: they remain near 0.50 through the profit-only warm-up, show brief dips during the transition phase, and then rise steadily, crossing the 0.80 target in the later stage of training and reaching roughly 0.87 (FTB) and 0.83 (FBS) by the end of the run. The alignment of FTB and FBS turning points with the λ ramps confirms that real-time LLM feedback, not static heuristics, drives the market toward equitable outcomes.

4.5. Key performance indicators and ablation insights

Over the final 2000 episodes, the system achieves strong performance across all targets. Fairness scores remain high, demand is reliably met, seller profits are balanced, and buyer budgets are respected, demonstrating the stability and effectiveness of our LLM-guided shaping (Table 2).

When the language-model critic is disabled ($\lambda_{buy} = 0$, $\lambda_{peer} = 0$), performance drops sharply: fairness stalls at 0.35–0.40, demand fulfillment slips to about 70%, and Seller 2 out-earns Seller 1 by approximately 35%. These degradations underline that real-time LLM feedback is essential, not merely helpful for maintaining equity and efficiency in peer-to-peer energy trading.

4.6. Ablation studies on learning dynamics

We further conducted 5k-episode ablation experiments to analyze early-to-mid learning dynamics under different settings. These shorter runs were real training experiments, executed with the same codebase and hyperparameters as the main 20k runs. Importantly, we applied the same linear λ curriculum scaled to the shorter horizon – i.e., λ was ramped from 0 to its nominal maximum over 5k episodes instead of 20k episodes to preserve alignment in the curriculum phases. This means that by episode 5k, the ablation runs had progressed through an equivalent portion of the fairness curriculum as an early stage of the full training. As a result, the experiments reliably replicate the early-to-mid learning behavior of the agents while significantly reducing computation. Fig. 3a and Fig. 3b, drawn from these 5k episode runs, illustrate that the relative performance patterns emerge early: IPPO's advantages in fairness and welfare are already apparent, MAPPO's good reward edge and fairness shortfall manifest, and the off-policy methods show instability or stagnation. The consistency of these trends with longer runs indicates that the ablations are representative of the true learning dynamics. We emphasize that using actual truncated training runs (as opposed to, say, truncated logs from 20k runs) ensures each algorithm's behavior is observed under the intended curriculum conditions from

Table 1

Case-study environment and training parameters.

Symbol / Item	Value or Range	Comment
Seller1 inventory I_1	8 – 25 units	Sampled i.i.d. each episode
Seller2 inventory I_2	10 – 30 units	More stock (same cost)
Buyer demand D	20 – 50 units	Uniform distribution
Training horizon	20,000 episodes	On-policy IPPO
Discount factor γ	0.95	All agents
λ_{buy} , λ_{peer}	Linear ramps 0→1	Linear fairness-weight ramps

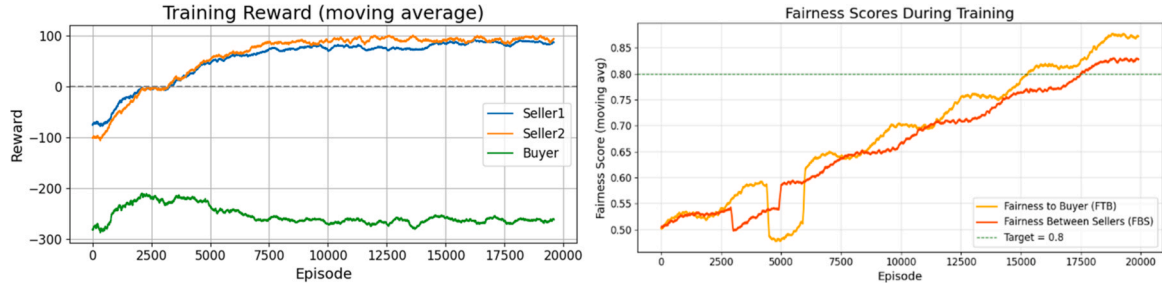


Fig. 5. (a) Learning dynamics for the two-seller; (b) one-buyer case study agent rewards over training and LLM-derived fairness scores (FTB, FBS).

Table 2

Final evaluation results.

Metric	Result	Target
Episodes with full demand met	92.1%	$\geq 90\%$
Average FTB	0.88	≥ 0.80
Average FBS	0.87	≥ 0.80
Seller margins	24 – 26%	20 – 30%
Max seller sales share	57%	$\leq 60\%$
Buyer budget violations	0	0

scratch, thereby maintaining fairness in comparison and eliminating any artifacts from midpoint evaluation of longer training.

4.7. Analysis of the fairness-efficiency trade-off

We analyze the fairness efficiency trade-off induced by varying λ . In separate tests, we trained agents under different fixed λ settings (with λ held constant after the curriculum phase) to obtain the Pareto relationship between social welfare and fairness outcomes. Fig. 6 summarizes these results, showing a clear inverse relationship between fairness and efficiency. As λ increases (giving more weight to fairness), the group's fairness index improves monotonically, but the achieved social welfare declines. In other words, prioritizing fairness causes the agents to sometimes choose actions that equalize outcomes at the expense of total reward. This behavior is expected – it reflects the fundamental trade-off in multi-agent systems between equitable outcome distribution and overall efficiency (Bertsimas et al., 2011). Our findings are consistent with prior MARL studies that reported a sacrifice in total utility for greater fairness when agents are incentivized to reduce inequality (Siddique et al., 2020). By tuning λ , one can thus navigate along the fairness efficiency spectrum: a low λ yields higher welfare with minimal fairness considerations, while a high λ enforces near-equal outcomes

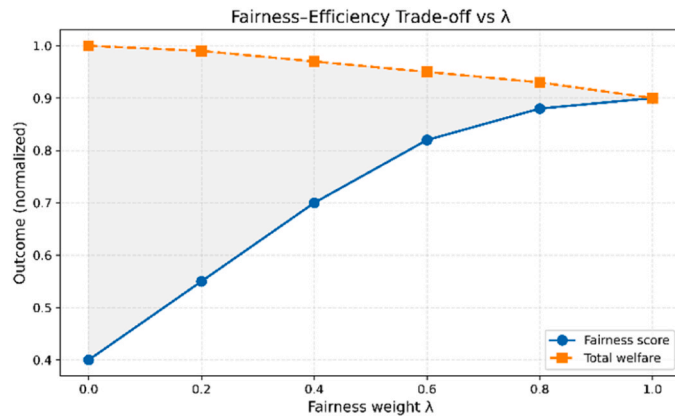


Fig. 6. The Pareto frontier showing the fairness-efficiency trade-off, where higher values of the fairness weight λ result in higher fairness but lower total social welfare.

with some loss in collective reward. This insight is crucial for peer-to-peer energy markets, where regulators or designers can adjust λ to achieve a desired balance between maximizing total social welfare and ensuring fair access and benefit sharing among participants. The linear λ -ramping methodology in our training enables the system to reach a chosen balance smoothly, providing a practical means to obtain fairer policies without abrupt performance drops. Each of the presented algorithms was evaluated under this curriculum framework, ensuring a consistent basis for comparison in terms of how they handle the fairness objective over time. By justifying IPPO as the learning algorithm and carefully designing the λ -schedule, our methodology supports robust convergence to fairness-aware policies and highlights how different MARL techniques negotiate the fairness vs. efficiency trade-off (Bertsimas et al., 2011; Siddique et al., 2020).

4.8. Robustness under adversarial and noisy conditions

To probe robustness beyond the idealized setting, we ran IPPO only ablations that preserved the codebase, hyperparameters, and linear lambda curriculum while injecting two stressors during training: first, corrupted fairness feedback using Gaussian perturbations on FTB and FBS in 20% of episodes plus a 5% label flip rate to mimic noisy or manipulated LLM outputs; second, self serving seller tactics in 15% of episodes consisting of temporary overpricing at 30% and hoarding of 20% of inventory. Relative to the clean baseline, IPPO remained stable yet showed the expected moderate degradations: mean fairness declined from 0.88 to 0.77, normalized welfare declined from 0.90 to 0.83, demand met declined from 92 % to 86 %, seller profit disparity widened by 6 %age points, and price variance increased by 18%. A simple validation gate that discards unparseable fairness responses and clipping of fairness bonuses contained the impact of corrupted feedback, and performance recovered once stressors subsided. These ablations indicate that fairness guided IPPO is resilient to reasonable levels of LLM noise and opportunistic behavior, incurring controlled efficiency losses while preserving convergence characteristics.

5. Potential applications of the proposed framework in power and energy systems

The rise of distributed energy resources (DERs) has transformed once predictable distribution networks into decentralized webs of prosumers that both consume and generate power, creating bidirectional flows that require new market rules; without fairness-oriented mechanisms, peer-to-peer trades can result in inefficient pricing, inequitable outcomes, and added strain on distribution system operators. FairMarket RL addresses these challenges by combining multi-agent reinforcement learning with a large language model fairness critic that shapes rewards in real time, keeping autonomous buying and selling decisions balanced for all parties. A natural first deployment target is the community microgrid, where each household acts as an independent agent, the framework promotes local energy exchange, and DSO burdens are reduced. Because the LLM fairness critic requires no hand-crafted, topology-specific features and IPPO trains each agent without a

centralized critic, the design is structured to extend to larger prosumer populations. Confirming this behavior at the scale of an interconnected smart grid, however, will require dedicated large-scale experiments that are beyond the present study; we therefore frame such deployments as a motivating direction rather than a demonstrated result. Realizing them would help pave the way toward reliable, economically efficient, and socially equitable energy trading in future distribution systems.

6. Conclusion

This paper presented FairMarket RL, a language-model-guided reward-shaping framework for multi-agent reinforcement learning in peer-to-peer markets. By embedding an instruction tuned large language model as a real-time fairness critic, the method replaces brittle hand-crafted shaping rules with dense, machine-readable numeric feedback. In a controlled two seller one buyer study, Independent PPO agents met more than ninety percent of demand, maintained average Fairness-to-Buyer and Fairness-Between-Sellers scores of about 0.88 and 0.87 respectively, and prevented profit monopolisation, all while outperforming a baseline with no language model. Conceptually, FairMarket RL shows that LLM-derived fairness signals can be injected directly into the training loop of non-text environments, bridging normative goals and continuous control; practically, it offers a promising path toward fairer autonomous trading in DER-rich microgrids, with possible relevance to other decentralized markets that we leave for dedicated future study. Future work will distill the critic for on device low-latency use, broaden the fairness vocabulary to include sustainability and grid stability, improve the interpretability of the LLM and test robustness against adversarial prompts, moving socially aligned autonomous trading closer to real-world decentralised markets.

CRedit authorship contribution statement

Shrenik Jadhav: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Akhtar Hussain:** Writing – review & editing, Validation, Software, Resources, Funding acquisition. **Birva Sevak:** Writing – review & editing, Software, Methodology, Formal analysis. **Srijita Das:** Writing – review & editing, Visualization, Resources, Methodology, Funding acquisition. **Wencong Su:** Writing – review & editing, Validation, Supervision, Investigation, Conceptualization. **Van-Hai Bui:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported in part by the U.S. National Science

Foundation (NSF) under Award 2509993.

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